



Raspberry Pi Pico W — CircuitPython Setup and Adhyay1 Library Guide

This file provides a complete step-by-step guide to set up Raspberry Pi Pico W with CircuitPython and use the Adhyay1 library — from installing CircuitPython, adding libraries, to running example code and troubleshooting.

1. Quick Introduction / Requirements

Hardware:

- Raspberry Pi Pico W
- USB Micro-B or Micro-USB to USB-A cable (data supported)- PC (Windows / macOS / Linux)

Software:

- [CircuitPython](#)
- Adhyay1 Library (downloadable as .mpy file or zip)
- Optional: Mu Editor (or Thonny IDE) for easy code editing and uploading

2. Install CircuitPython on Pico W

1. Connect the Pico W to your PC while holding the BOOTSEL button.
2. It will appear as a USB drive named RPI-RP2.
3. Download the latest CircuitPython UF2 for Pico W from [CircuitPython Downloads](#).
4. Drag and drop the .uf2 file to the RPI-RP2 drive.
5. The board will reboot, and a new drive CIRCUITPY should appear.

3. Add Adhyay1 Library

1. Download the Adhyay1 library from your repository or release zip.
2. Copy the Adhyay1.mpy (or library folder) into the lib folder inside CIRCUITPY. If lib folder does not exist, create it.
3. Your Pico W is now ready to use Adhyay1 functions.

4. Writing and Running Your First Script

1. Open Thonny IDE or Mu Editor.
2. Select Python (CircuitPython) as the interpreter.
3. Create a new script, for example code.py.

4. Example using Adhyay1 for LED blink:

```
import time
from adhyay1 import led
# Initialize onboard LED
led_pin = 25
while True:
    led.on(led_pin)
    time.sleep(1)
    led.off(led_pin)
    time.sleep(1)
```

5. Save the script to the root of CIRCUITPY. It will run automatically.

5. Using Serial Console

- Open Thonny → View → Serial or Mu Editor → REPL.
- Use print() statements to debug and monitor outputs.

Example:

```
import time
while True:
    print("Hello from Adhyay1")
    time.sleep(1)
```

6. Installing Additional Libraries (Optional)

- Use [CircuitPython Bundle](#) to download dependencies like adafruit_dht, adafruit_motor, etc.
- Copy them into the lib folder on CIRCUITPY.

7. Common Troubleshooting

- CIRCUITPY not appearing: Check cable is data-capable and reflash UF2.
- Import errors: Ensure Adhyay1.mpy is inside lib folder.
- Script not running: Make sure file is named code.py and saved in root of CIRCUITPY.
- Permission issues (Linux): Mounting should be automatic, but try using a different USB port if needed.

Your Raspberry Pi Pico W is now ready for CircuitPython coding with the Adhyay1 library!